

BAKR ALHAYEK, M.D.

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EDUCATION

Doctor of Medicine (M.D.), Hama University Faculty of Medicine — Hama, Syria

2022

Graduated with Excellence (GPA 4.4/5.0; U.S. equivalent 3.79/4.0). ECFMG certified, July 2023.

POSTGRADUATE TRAINING

Internal Medicine Residency, AdventHealth Tampa — Tampa, FL

July 2024 – June 2027

Selected as Chief Medical Resident (Research & Innovation), 2026–2027. Board-eligible in Internal Medicine, 2027.

LICENSURE & BOARD SCORES

USMLE — Step 1: 244 • Step 2 CK: 251 • Step 3: 246 (all passed on first attempt)

ECFMG Certificate (July 2023) • Florida Medical License (application pending)

CLINICAL EXPERIENCE

Inpatient Hospital Medicine, AdventHealth Tampa — Tampa, FL

July 2024 – Present

High-acuity general medicine teaching service (census 18) carrying senior-resident, attending-level responsibility for complex, multi-comorbid patients. Focused depth in inpatient oncology and oncologic emergencies — neutropenic fever, tumor lysis syndrome, malignant hypercalcemia — and immune checkpoint inhibitor toxicities (myocarditis, pneumonitis, colitis), managed within multidisciplinary IM–Oncology teams. Daily Epic use.

Internship, General Internal Medicine, Damascus University Hospital — Damascus, Syria

Mar – Oct 2022

RESEARCH EXPERIENCE

Research Fellow, Hematology/Oncology, Mayo Clinic — Jacksonville, FL

Nov 2022 – Nov 2023

Mentor: Muhamad Alhaj Moustafa, MD, MS. Outcomes research in hematologic malignancies and solid tumors; multiple first-author manuscripts (AJCO, Cureus, Respiratory Case Reports).

Clinical Research Associate, Hematology/Oncology, USF Health Morsani College of Medicine — Tampa, FL

Nov 2023 –

Jun 2024

PIs: Bradley Fletcher, MD, PhD; Nathan Visweshwar, MD. Clinical research in hematologic malignancies; first-author manuscripts in AJCO with collaborators at UAB and the University of Alberta.

MANUSCRIPTS UNDER REVIEW / IN PREPARATION

Alhayek B, et al. Electrophysiologic Phenotype Outperforms LVEF for Mortality Risk Stratification in ICI-Associated Myocarditis: A Systematic Review and Meta-Analysis of 319 Cases. Submitted, JACC: Advances. [First Author]

Alhayek B, Baidoun F, et al. Impact of Peri-Operative Chemotherapy on Outcomes in Non-Metastatic Biliary Tract Cancer. Preprint posted Aug 2025; under journal review. [First Author]

Alhayek B, et al. Baseline Evaluation of Inappropriate HIT Testing to Inform Diagnostic Stewardship. In preparation. [First Author, Project Lead]

PEER-REVIEWED PUBLICATIONS

11 PubMed-indexed publications (7 first-author). Reverse chronological; name in bold.

2026

Alhayek B, Malone X, Alwan I, Sepulveda L, Kroker-Bode C, Ramsakal A. Change Through Learning: Educational Interventions to Improve Cancer Pain Management. J Natl Compr Canc Netw. 2026;24(3.5). PMID: 41990810. [First Author]

Malone X, **Alhayek B**, Khan A, Ramsakal A. Implementing an HPV Vaccination Program in an Internal Medicine Resident Ambulatory Care Clinic. J Natl Compr Canc Netw. 2026;24(3.5). PMID: 41990828.

2025

Alhayek B, Malone X, Rashid MA, Sepulveda L, Ramsakal A. Fulminant Immune Checkpoint Inhibitor-Induced Myocarditis and Complete Heart Block in Advanced Melanoma. Cureus. 2025;17(11):e98075. PMID: 41473611. [First Author]

Alhayek B, Malone X, Rabie T, Brink R, Gummalla R. Mixed-Phenotype Acute Leukemia Transforming Into Acute Myelomonocytic Leukemia (AML M4). *Cureus*. 2025;17(11). PMID: 41393639. [First Author]

Alhayek B, Malone X, Mislay JA, Cardew RT, Ramsakal A. Epstein-Barr Virus-Induced Hemophagocytic Lymphohistiocytosis in Adolescence. *Cureus*. 2025;17(10). PMID: 41262800. [First Author]

Alhayek B, Baidoun F, Hadidi D, Alhaj Moustafa M, Abdel-Rahman O. Impact of Intensive Multimodal Treatment on the Outcomes of Patients With Anaplastic Thyroid Cancer. *Am J Clin Oncol*. 2025;49(3):136-146. PMID: 40772694. [First Author]

Jlailati A, Al Sbenaty G, Boali O, Younes D, **Alhayek B**, et al. Radiation-Induced Temporal Lobe Necrosis After External Beam Radiotherapy for Nasopharyngeal Cancer. *J Med Case Rep*. 2025;19(1):200. PMID: 40652209.

Alhayek B, Malone X, Khan A, Gummalla R, Brink R. Pancreatic Plasmacytoma Presenting as Acute Pancreatitis: An Unusual Extramedullary Onset of Multiple Myeloma. *Cureus*. 2025;17(6). PMID: 40621238. [First Author]

Alhayek B, Baidoun F, Hadidi D, Moustafa MA, Abdel-Rahman O. Impact of Sequencing of Treatment Modalities on Survival in Nonmetastatic Hepatocellular Carcinoma. *Am J Clin Oncol*. 2025;48(12):600-609. PMID: 40476702. [First Author]

Malone X, **Alhayek B**, Telleria O, Garza JF. Spontaneous Tumor Lysis Syndrome With Acute Kidney Injury and the Timing of Chemotherapy. *Cureus*. 2025;17(6). PMID: 40621233.

2023

Wadiwala IJ, Yu Lee-Mateus A, **Alhayek B**, et al. Atypical Distal Tracheal Fibrous Bridge and Bronchial Stenosis in an Adult With Bronchopulmonary Dysplasia. *Respirol Case Rep*. 2023;11(8):e01184. PMID: 37593371.

CONFERENCE ABSTRACTS & PRESENTATIONS

2026

Alhayek B, Abubakir M, Potti S, Abdel-Rahman OM. State-Level Trends in Colorectal Cancer Incidence and Stage Before and During the COVID-19 Pandemic. *ASCO Annual Meeting 2026*. [First Author, Poster]

Alhayek B, Potti S, Behera R, Sonpavde GP, et al. Early Neutrophil-to-Lymphocyte Ratio Dynamics to Predict Primary Refractory Disease in Metastatic Urothelial Carcinoma on PD1/L1 Inhibitors. *GU ASCO 2026*. [First Author, Poster]

Alhayek B, Potti S, Behera R, Sonpavde GP, et al. Any Regression of Tumor (ART) to Discriminate Overall Survival in Metastatic Urothelial Carcinoma With Stable Disease by RECIST 1.1. *GU ASCO 2026*. [First Author, Poster]

Elhusseiny KM, **Alhayek B**, Hajjar AH, et al. Primary Squamous Cell Carcinoma of the Small Intestine: A Population-Based Study. *ASCO GI 2026*. [Co-Author, Poster]

Alswaiti Y, Alzibdeh A, Shahwan G, et al., **Alhayek B**. Prediction of Lymphovascular Invasion in Breast Carcinoma Using MRI-Based Radiomics: A Systematic Review and Meta-Analysis. *ASCO Annual Meeting 2026*. [Co-Author]

2025

Khan A, **Alhayek B**, Harris MD. Hepatic Mucinous Cystic Neoplasm Causing Biliary Obstruction. *ACG Annual Meeting 2025*. [Co-Author, Case Report]

2024

Alhayek B, Malone X, Alwan I, Sepulveda L, Kroker-Bode C, Ramsakal A. Educational Interventions to Improve Cancer Pain Management. *JNCCN Annual Conference 2024*. [Lead Author]

Malone X, **Alhayek B**, Khan A, Ramsakal A. Implementing an HPV Vaccination Program in an IM Resident Clinic. *JNCCN Annual Conference 2024*. [Co-Author]

Alhayek B, et al. Fulminant Immune Checkpoint Inhibitor-Induced Myocarditis and Complete Heart Block. *JNCCN Annual Conference 2024*. [First Author]

et al., **Alhayek B**. Rare Case of Metastatic Lung Cancer Presentation. *JNCCN Annual Conference 2024*. [Co-Author]

QUALITY IMPROVEMENT & PATIENT SAFETY

Diagnostic Stewardship — HIT Testing (Project Lead), AdventHealth Tampa 2024 – Present
Retrospective review of ~200 PF4 assays adjudicated by 4T score; ~70% inappropriate ordering identified, driving an Epic Best Practice Alert projected to save ~\$17,000/year.

Opioid Stewardship — Cancer Pain Curriculum (Lead), AdventHealth Tampa 2024 – Present
Multidisciplinary resident curriculum in cancer pain management; knowledge scores improved 62% → 82% (p<0.001, d=0.96). Published, *JNCCN* 2026.

Cancer Prevention — HPV Vaccination Outreach (Co-Lead), AdventHealth Tampa 2024
Epic SlicerDicer–driven outreach to 240 eligible patients; 74% reached, 35% vaccination acceptance. Published, *JNCCN* 2026.

Health Equity — Uninsured ED Hypertension Pathway (Co-Lead), AdventHealth Tampa 2024 – Present

ED-to-continuity pathway (point-of-care education, home BP cuffs, 7-day follow-up); targeting a 20% reduction in 30-day ED revisits. Data collection ongoing.

LEADERSHIP & TEACHING

Chief Medical Resident — Research & Innovation, AdventHealth Tampa

July 2026 – June 2027

Selected by program leadership to lead the resident scholarly portfolio, mentor junior residents in research and QI, and build the program's research infrastructure.

Clinical Artificial Intelligence Lead, AdventHealth Tampa

2025 – Present

Department AI lead partnering with the enterprise AI team on responsible clinical AI at the bedside; lead recurring “AI in Medicine” sessions on informatics, large language model applications, and AI-assisted clinical decision support.

Founding Member, Scholarly Activity Committee (AHT-SAC), AdventHealth Tampa

2024 – Present

Founded the internal medicine resident research committee as a PGY-1; now the program's primary research coordination mechanism.

Author, A Practical Guide for Resident Research, AdventHealth Tampa

2024

Authored a 50-page evidence-based handbook with plans to be adopted as the official research curriculum for the IM residency.

PROFESSIONAL MEMBERSHIPS

American College of Physicians (ACP) • Society of Hospital Medicine (SHM), Oncology SIG • American Society of Clinical Oncology (ASCO)

SKILLS

Clinical / Data: Epic (SlicerDicer for QI and outreach); SPSS, REDCap, RStudio, BlueSky Statistics.

Clinical AI: enterprise clinical AI implementation and physician education (Clinical AI Lead); LLM applications, AI-assisted review tools (Rayyan, Covidence), and responsible AI governance in healthcare.

Languages: Arabic (native), English (fluent).

VOLUNTEER & COMMUNITY SERVICE

Volunteer Physician, Community Health Outreach, AdventHealth Tampa

2024 – Present

5+ community outreach events providing health counseling, point-of-care testing, and care navigation for underserved Tampa-area residents.

Volunteer Clinical Assistant, M.A.S.S. Free Clinic — Jacksonville, FL

2023 – Present

Support intake, clinical histories, patient education, and outreach for uninsured and low-income patients.

REFERENCES

Claudia Kroker-Bode, M.D., Ph.D.

Program Director, Internal Medicine Residency, AdventHealth Tampa • claudia.krokerbode.md@adventhealth.com

Guru P. Sonpavde, M.D.

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Lidia Sepulveda Rubiera, M.D.

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